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Group Project in Ergodic Theory

Weak Mixing Implies Weak Mixing of All Orders Along Multiples

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Introduction

The report summarizes certain results in [2], where Furstenberg introduced an ergodic-theoretic approach to prove Szemerédi's Theorem on arithmetic progressions. His method used ergodic theory and did not rely on the van der Waerden, Roth, or Szemerédi results. The ergodic approach has since been generalized to more complex systems.

In the report, we follow the foundational work of [2], which is also presented in [3], to establish Furstenberg's result for weak mixing transformations, that is, if $T : X \to X$ is a weak mixing measure-preserving transformation on a probability space $(X, \mathcal{B}, \mu)$, and $A_1, A_2, \ldots, A_k \in \mathcal{B}$, then

$$\lim_{N-M\to\infty} \frac{1}{N-M} \sum_{n=M+1}^{N} |\mu(A_1 \cap T^n A_2 \cap \cdots \cap T^{(k-1)n} A_k) - \mu(A_1)\mu(A_2)\ldots\mu(A_k)| = 0$$

which may be regarded as the statement that T is weak mixing of all orders along multiples.

1 Preliminaries

In this section we first set out the notation that will be used in the sequel and then present basic results regarding measure-preserving transformations, ergodicity and weak mixing. As usual, we write $L^p(X, \mathcal{B}, \mu)$ for the space of measurable functions with $(\int |f|^p d\mu)^{1/p} < \infty$ and $L^\infty(X, \mathcal{B}, \mu)$ for the space of essentially bounded measurable functions.

Definition 1.1 (Measure-preserving transformation)**.** A transformation $T : (X, \mathcal{B}, \mu) \to (X, \mathcal{B}, \mu)$ of a measure space onto itself is called a *measure-preserving transformation* if for any measurable set B, then full preimage $T^{-1}(B)$ is also measurable and $\mu(T^{-1}(B)) = \mu(B)$. In this case, we call $(X, \mathcal{B}, \mu, T)$ a *measure-preserving system.*

Remark 1.2. Let $T : (X, \mathcal{B}, \mu) \to (X, \mathcal{B}, \mu)$ be a measure-preserving transformation. Then T determines a unitary operator $U_T : L^2(X, \mathcal{B}, \mu) \to L^2(X, \mathcal{B}, \mu)$ defined by $U_T f = f \circ T$. For now, we will denote this map by Tf.

Proposition 1.3 (Change of variables)**.** *If $\phi : (X, \mathcal{B}, \mu) \to (Y, \mathcal{C})$ is a measurable map and f is a complex-valued measurable function on Y, then*

$$\int_Y f\, d(\mu\phi^{-1}) = \int_X f \circ \phi \, d\mu.$$

Proposition 1.3 immediately yields the following consequence.

Proposition 1.4. *Let $T : (X, \mathcal{B}, \mu) \to (X, \mathcal{B}, \mu)$ be a measure-preserving transformation. Then*

$$\int f \ d\mu = \int Tf \ d\mu.$$

Definition 1.5 (Ergodicity)**.** A measure μ is said to be *ergodic* with respect to T if for any measurable $A \subset X$ with $T^{-1}(A) = A$, either $\mu(A) = 0$ or $\mu(X \setminus A) = 0$.

Definition 1.6 (Weak mixing)**.** Let $T : (X, \mathcal{B}, \mu) \to (X, \mathcal{B}, \mu)$ be a measure-preserving transformation. Then T is *weak mixing* if for any two measurable sets A, B

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \left|\mu(T^{-1}A \cap B) - \mu(A)\mu(B)\right| = 0.$$

Proposition 1.7. *Let $T : X \to X$ be a measure-preserving transformation on a measure space $(X, \mathcal{B}, \mu)$. Then the following are equivalent:*

(1) The system $(X, \mathcal{B}, \mu, T)$ is weak mixing.

(2) $T \times S$ is ergodic on $X \times Y$ for each ergodic $(Y, \mathcal{C}, \nu, S)$.

(3) For any two measurable sets A, B, there exists a set $E \subset \mathbb{N}$ of density 0 such that

$$\lim_{\substack{n\to\infty \\ n \notin E}} \mu(T^{-n}A \cap B) = \mu(A)\mu(B).$$

Proposition 1.8. *Every weak mixing transformation is ergodic.*

For $n \geq 1$, the nth *iterate* of a map $T : X \to X$ is the n-fold composition $T^n = T \circ \cdots \circ T$.

Proposition 1.9. *If $T : X \to X$ is weak mixing, then $T^n : X \to X$ is weak mixing.*

Proof. Let A and B be measurable sets and let $a_k = \mu(T^{-k}A \cap B)$. By Proposition 1.7 (3), we can find a set $E \subset \mathbb{N}$ with density 0 so that the sequence $(a_k)_{k \in \mathbb{N} \setminus E}$ converges to $\mu(A)\mu(B)$. In particular, the subsequence $(a_{kn})_{k \in \mathbb{N} \setminus E*}$ converges to $\mu(A)\mu(B)$, where $E^* = \{\ell \in \mathbb{N} \mid \ell n \in E\}$. In other words,

$$\lim_{\substack{k \to \infty \\ k \notin E^*}} \mu(T^{-kn}A \cap B) = \mu(A)\mu(B).$$

To complete the proof we now show that E^* has density 0, that is,

$$\lim_{k \to \infty} \frac{|E^* \cap \{1, \ldots, k\}|}{k} = 0.$$

Since E has density 0, it suffices to verify that $|E^* \cap \{1, \ldots, k\}| \leq |E \cap \{1, \ldots, kn\}|$ for any k. Let k be a fixed integer. For each $\ell \in E^* \cap \{1, \ldots, k\}$, we have $\ell n \in E$. Since $\ell \in \{1, \ldots, k\}$, we get $\ell n \in \{n, 2n \ldots, kn\} \subset \{1, 2 \ldots, kn\}$. So $\ell n \in E \cap \{1, 2 \ldots, kn\}$. This implies $|E^* \cap \{1, \ldots, k\}| \leq |E \cap \{1, \ldots, kn\}|$, as desired. □

Proposition 1.10. *If $T : X \to X$ is weak mixing, then $T_k = T \times T^2 \times \cdots \times T^k$ is ergodic on $X_k = X \times \cdots \times X$ for each $k \geq 1$.*

Proof. In view of Propositions 1.8 and 1.9, (X, T^j) is ergodic for each $j \geq 1$. Now we prove by induction on k. Suppose $k > 1$ and (X_k, T_k) is ergodic. Since (X, T^{k+1}) is ergodic, it follows from Proposition 1.7 (2) that $(X_{k+1}, T_{k+1}) = (X_k \times X, T_k \times T^{k+1})$ is ergodic, completing the induction. □

We end this section by stating two fundamental theorems in ergodic theory.

Theorem 1.11 (Mean Ergodic Theorem). *If $T : (X, \mu) \to (X, \mu)$ is a measure-preserving transformation of a measure space and $f \in L^2(X, \mu)$, then*

$$\frac{1}{n} \sum_{i=0}^{n-1} f \circ T^i \to P_T(f)$$

in $L^2(X, \mu)$ as $n \to \infty$, where P_T is the orthogonal projection to the subspace of L^2 T-invariant functions.

Remark 1.12. If the system $(X, \mathcal{B}, \mu, T)$ is ergodic, then the limit equals to $\int f \, d\mu$.

Theorem 1.13 (Birkhoff Pointwise Ergodic Theorem). *If $T : X \to X$ is an ergodic μ-preserving transformation, $\mu(X) = 1$, and $\phi \in L^1(X, \mu)$, then for every x outside of a set of measure zero,*

$$\lim_{n \to \infty} \frac{1}{n} \sum_{k=0}^{n-1} \phi(T^k(x)) = \int \phi \, d\mu.$$

2 Generic Measure

Definition 2.1. Let $T : X \to X$ be an ergodic measure-preserving transformation on a finite measure space $(X, \mathcal{B}, \mu)$ and let ν be another measure on $(X, \mathcal{B})$. Let $\mathcal{A} \subset L^\infty(X, \mathcal{B}, \mu)$ be a T-invariant (that is $T\mathcal{A} = \mathcal{A}$) self-conjugate algebra of bounded measurable functions on X, and let $\{M_k, N_k\}$ be a sequence of pairs of integers with $N_k - M_k \to \infty$ as $k \to \infty$. We say that ν is *generic* for μ with respect to $\mathcal{A}$ and $\{M_k, N_k\}$ if for all $f \in \mathcal{A}$,

$$\frac{1}{N_k - M_k} \sum_{n=M_k+1}^{N_k} \int f(T^n x) \, d\nu(x) \to \int f \, d\mu$$

as $k \to \infty$.

Let $k \geq 2$ and $(X, \mathcal{B}, \mu)$ be a measure space. We write $(X_k, \mathcal{B}_k, \mu_k) = (X \times \cdots \times X, \mathcal{B} \times \cdots \times \mathcal{B}, \mu \times \cdots \times \mu)$, where each of the products being taken k times. Let $x \in X$ and $\mathcal{S}$ be a σ-algebra on X. Recall that the Dirac measure δ_x on $(X, \mathcal{S})$ is defined by

$$\delta_x(S) = \begin{cases} 1 & \text{if } x \in S, \\ 0 & \text{if } x \notin S. \end{cases}$$

We present several properties of genericity among measures, one of which will serve as the base case in the proof of Theorem 4.1.

Proposition 2.2. *Let $(X, \mathcal{B}, \mu, T)$ and $\mathcal{A}$ be defined as in Definition 2.1. Then the following properties hold.*

(1) μ is always generic for μ.

(2) If $\mathcal{B}$ is countably generated with respect to μ, then for almost all x, the Dirac measure δ_x is generic for μ with respect to $\mathcal{A}$ and $\{0, k\}$.

(3) The diagonal measure ν_2 on $(X_2, \mathcal{B}_2)$ defined by

$$\int f(x_1, x_2) \, d\nu_2(x_1, x_2) = \int f(x, x) \, d\mu(x) \quad (f \in L^\infty(X_2, \mathcal{B}_2))$$

is generic for μ_2 with respect to any sequence $\{M_k, N_k\}$ for which $N_k - M_k \to \infty$ and the algebra

$$\mathcal{A}_2 = \left\{ \sum_{i=1}^{n} f_i(x_1) g_i(x_2) : f_i, g_i \in L^\infty(X, \mathcal{B}, \mu) \right\}.$$

We denote such a function of the above form as $\sum_{i=1}^{n} f_i \otimes g_i$.

Proof. (1) Since T is a measure-preserving transformation, it follows from Proposition 1.4 that $\int f(T^n x) \, d\mu(x) = \int f d\mu$ for any n. This proves our assertion.

(2) We wish to construct a set $E \in \mathcal{B}$ so that $\mu(E) = 0$ and for every $x \in X \setminus E$,

$$\frac{1}{k}\sum_{n=1}^{k}\int T^n f \; d\delta_x \to \int f \; d\mu$$

for all $f \in \mathcal{A}$. Since $\int f \circ T^n \; d\delta_x = f(T^n x)$ for any n, it suffices to show that for every $x \in X \setminus E$,

$$\frac{1}{k}\sum_{n=1}^{k} f(T^n x) \to \int f \; d\mu$$

for all $f \in \mathcal{A}$.

Note that $\Omega \subset \mathcal{A} \subset L^1(X, \mathcal{B}, \mu)$. Since $\mathcal{B}$ is countably generated with respect to μ, we see that the subspace $\mathcal{A}$ of $L^1(X, \mathcal{B}, \mu)$ is separable. Hence we can find a countable subset $\Omega = \{f_i \,|\, i \in \mathbb{N}\}$ of $\mathcal{A}$ whose closure equals $\mathcal{A}$, in other words, Ω is dense in $\mathcal{A}$ with respect to $L^1(X, \mathcal{B}, \mu)$ norm. Using Pointwise Ergodic Theorem (Theorem 1.13), one sees immediately that for each $f_i \in \Omega$, there exists a set $E_i \in \mathcal{B}$ with $\mu(E_i) = 0$ such that for every $x \in X \setminus E_i$,

$$\frac{1}{k}\sum_{n=1}^{k} f_i(T^n x) \to \int f_i \; d\mu.$$

For $i, j \in \mathbb{N}$, we let $E_{i,j}$ be the set in $\mathcal{B}$ such that for every $x \in X \setminus E_{i,j}$,

$$\frac{1}{k}\sum_{n=1}^{k} |f_i - f_j|(T^n x) \to \int |f_i - f_j| \; d\mu.$$

Now we construct the set E with the desired properties. Let

$$E = \bigcup_{i \in \mathbb{N}} E_i \cup \bigcup_{i,j \in \mathbb{N}} E_{i,j}.$$

Clearly $\mu(E) = 0$. Let $x \in X \setminus E$ and let $f \in \mathcal{A}$. Let $\varepsilon > 0$. Because Ω is dense in $\mathcal{A}$, there exist $f_\ell, f_m \in \Omega$ such that $\|f - f_\ell\|_1 < \varepsilon$ and $\|f_m - f_\ell\|_1 < \varepsilon$. By the Triangle Inequality, we have

$$\begin{aligned}\left|\frac{1}{k}\sum_{n=1}^{k} f(T^n x) - \int f \; d\mu\right| \leq{}& \left|\frac{1}{k}\sum_{n=1}^{k} |f - f_\ell|(T^n x)\right| \\ &+ \left|\frac{1}{k}\sum_{n=1}^{k} f_\ell(T^n x) - \int f_\ell \; d\mu\right| + \left|\int f_\ell \; d\mu - \int f \; d\mu\right|.\end{aligned}$$

Since $\frac{1}{k}\sum_{n=1}^{k} f_\ell(T^n x) \to \int f_\ell \; d\mu$ and $|\int f_\ell \; d\mu - \int f \; d\mu| < \varepsilon$,

$$\limsup_{k\to\infty}\left|\frac{1}{k}\sum_{n=1}^{k} f(T^n x) - \int f \; d\mu\right| \leq \limsup_{k\to\infty}\frac{1}{k}\sum_{n=1}^{k} |f - f_\ell|(T^n x) + \varepsilon.$$

Note that $||f - f_m| - |f_m - f_\ell|| \le |f - f_\ell| < \varepsilon$. Since ε is arbitrary small, $|f - f_m|$ is close to $|f_m - f_\ell|$. Hence $\limsup_{k\to\infty} \frac{1}{k}\sum_{n=1}^{k} |f - f_\ell|(T^n x) = \limsup_{k\to\infty} \frac{1}{k}\sum_{n=1}^{k} |f_m - f_\ell|(T^n x) = \int |f_m - f_\ell| \, d\mu \le \varepsilon$. Therefore

$$\limsup_{k\to\infty} \left| \frac{1}{k} \sum_{n=1}^{k} f(T^n x) - \int f \, d\mu \right| \le 2\varepsilon.$$

So δ_x is generic for μ with respect to $\mathcal{A}$ and $\{0, k\}$.

(3) By the Mean Ergodic Theorem, if $f \in L^2(X, \mathcal{B}, \mu)$, then

$$\frac{1}{N_k - M_k}(Tf + T^2 f + \cdots + T^{N_k - M_k} f) \to \int f \, d\mu$$

in $L^2(X, \mathcal{B}, \mu)$. Using Proposition 1.4, we see that

$$\frac{1}{N_k - M_k} \sum_{j=M_k+1}^{N_k} T^j f \to \int f \, d\mu$$

in $L^2(X, \mathcal{B}, \mu)$. This implies that if $f, g \in L^\infty(X, \mathcal{B}, \mu) \subset L^2(X, \mathcal{B}, \mu)$, then

$$\left\langle g, \frac{1}{N_k - M_k} \sum_{j=M_k+1}^{N_k} T^j f \right\rangle \to \left\langle g, \int f \, d\mu \right\rangle$$

where $\langle \cdot, \cdot \rangle$ is the standard inner product on $L^2(X, \mathcal{B}, \mu)$. Hence

$$\frac{1}{N_k - M_k} \sum_{j=M_k+1}^{N_k} \int g \cdot T^j f \, d\mu \to \int f \, d\mu \cdot \int g \, d\mu.$$

Replacing x by $T^n x$ gives

$$\frac{1}{N_k - M_k} \sum_{n=M_k+1}^{N_k} \int g(T^n x) f(T^{2n} x) d\mu \to \int f \, d\mu \cdot \int g \, d\mu,$$

or

$$\frac{1}{N_k - M_k} \sum_{n=M_k+1}^{N_k} \int T_2^n (g \otimes f) dv_2 \to \int g \otimes f \, d\mu_2$$

where $T_2^n = T^n \times T^{2n}$. □

3 Mean Ergodic Theorem for Generic Measures

We wish to generalize Proposition 2.2 (3) from $(X_2, \mathcal{B}_2, \mu_2, T_2)$ to $(X_k, \mathcal{B}_k, \mu_k, T_k)$ where $k \ge 2$. Before proving this, we need the Mean Ergodic Theorem for generic measures.

Theorem 3.1. *If $(X, \mathcal{B}, \mu, T)$ is ergodic and ν is generic for μ with respect to $\{M_k, N_k\}$ and an algebra $\mathcal{A}$ defined as in Definition 2.1, then for each $f \in \mathcal{A}$,*

$$\frac{1}{N_k - M_k} \sum_{n=M_k+1}^{N_k} T^n f \to \int f \, d\mu$$

in $L^2(X, \mathcal{B}, \nu)$ as $k \to \infty$. If we take $\nu = \mu$ we recover the ordinary Mean Ergodic Theorem.

Proof. Let $f \in \mathcal{A}$. Without loss of generality, assume that $\int f \, d\mu = 0$. Given $\varepsilon > 0$, we choose an integer Q so large that

$$\int \left| \frac{Tf + T^2 f + \cdots + T^Q f}{Q} \right|^2 d\mu < \varepsilon.$$

Since $\mathcal{A}$ is a T-invariant self-conjugate algebra, we have

$$g = \left| \frac{Tf + T^2 f + \cdots + T^Q f}{Q} \right|^2 \in \mathcal{A}.$$

By applying the definition of genericity to g, we see that

$$\frac{1}{N_k - M_k} \sum_{n=M_k+1}^{N_k} \int T^n g \, d\nu \to 0,$$

namely,

$$\frac{1}{N_k - M_k} \sum_{n=M_k+1}^{N_k} \int \left| \frac{T^{n+1} f + \cdots + T^{n+Q} f}{Q} \right|^2 d\nu < \varepsilon$$

for sufficiently large k. Let

$$A_{Q,k} = \frac{1}{(N_k - M_k)Q} \sum_{n=M_k+1}^{N_k} \sum_{i=1}^{Q} T^{n+i} f,$$

$$B_k = \frac{1}{N_k - M_k} \sum_{n=M_k+1}^{N_k} T^n f.$$

By the Cauchy-Schwarz Inequality, we have

$$|A_{Q,k}|^2 \leq \frac{1}{N_k - M_k} \sum_{n=M_k+1}^{N_k} \left| \frac{T^{n+1} f + \cdots + T^{n+Q} f}{Q} \right|^2.$$

It follows that $\int |A_{Q,k}|^2 d\nu < \varepsilon$. Now we claim that $A_{Q,k} - B_k \to 0$ in $L^2(X, \mathcal{B}, \nu)$ as $k \to \infty$.

Note that

$$\begin{aligned}A_{Q,k} - B_k &= \frac{1}{(N_k - M_k)Q} \left(\sum_{n=M_k+1}^{N_k} \sum_{j=1}^{Q} T^{n+j} f - Q \sum_{n=M_k+1}^{N_k} T^n f \right) \\ &= \frac{1}{(N_k - M_k)Q} \sum_{n=M_k+1}^{N_k} \left(\sum_{j=1}^{Q} T^{n+j} f - QT^n f \right) \\ &= \frac{1}{(N_k - M_k)Q} \sum_{n=M_k+1}^{N_k} \sum_{j=1}^{Q} \left(T^{n+j} f - T^n f \right) \\ &= \frac{1}{(N_k - M_k)Q} \sum_{j=1}^{Q} \sum_{n=M_k+1}^{N_k} \left(T^{n+j} f - T^n f \right) \\ &= \frac{1}{(N_k - M_k)Q} \sum_{j=1}^{Q} \left(\sum_{n=N_k-j+1}^{N_k} T^{n+j} f - \sum_{n=M_k+1}^{M_k+j} T^n f \right).\end{aligned}$$

Hence the $L^2(X, \mathcal{B}, \nu)$ norm of $A_{Q,k} - B_k$ is bounded by

$$\frac{2Q\|f\|_\infty}{N_k - M_k},$$

which converges to 0 as $k \to \infty$. Thus $A_{Q,k} - B_k \to 0$ in $L^2(X, \mathcal{B}, \nu)$ as $k \to \infty$. Consequently,

$$\limsup_{k\to\infty} \int |B_k|^2 \, d\nu \leq \varepsilon,$$

which proves the theorem. □

4 Weak Mixing of All Orders Along Multiples

Recall that $T_k = T \times T^2 \times \cdots \times T^k$. We shall now apply Theorem 3.1 to establish our main result. Unless otherwise stated, we assume that $T : (X, \mathcal{B}, \mu) \to (X, \mathcal{B}, \mu)$ is a measure-preserving transformation on a probability space $(X, \mathcal{B}, \mu)$.

Theorem 4.1. *If $(X, \mathcal{B}, \mu, T)$ is weak mixing, then for each $k \geq 2$ the diagonal measure ν_k on $(X_k, \mathcal{B}_k)$ defined by*

$$\int f(x_1, x_2, \ldots, x_k) \, d\nu_k = \int f(x, x, \ldots, x) \, d\mu(x) \quad (f \in L^\infty(X_k, \mathcal{B}_k))$$

is generic for μ_k with respect to T_k, any sequence $\{M_k, N_k\}$ with $N_k - M_k \to \infty$ as $k \to \infty$, and the algebra of all functions of the form

$$\sum_{j=1}^{J} f_1^j \otimes \cdots \otimes f_k^j(x_1, \ldots x_k) = \sum_{j=1}^{J} f_1^j(x_1) f_2^j(x_2) \ldots f_k^j(x_k) \quad (\textit{each } f_i^j \in L^\infty(X, \mathcal{B}, \mu)).$$

That is,

$$\frac{1}{N}\sum_{n=1}^{N}\int f_1(x)f_2(T^nx)\dots f_k(T^{(k-1)n}x)\ d\mu(x) \to \int f_1 d\mu\cdots\int f_k\ d\mu$$

for $f_1,\dots,f_k \in L^\infty(X,\mathcal{B},\mu)$.

Proof. The proof is by induction on k. By Proposition 2.2 (3), the statement holds for $k=2$. Assume now that the theorem is true for k, and we will prove it for $k+1$. Since $(X_k,\mathcal{B}_k,\mu_k,T_k)$ is ergodic by Propositions 1.8 and 1.9, it follows from Theorem 3.1 that, given $f_i \in L^\infty(X,\mathcal{B},\mu)$, $i=1,\dots,k$,

$$\frac{1}{N_k-M_k}\sum_{n=M_k+1}^{N_k} T_k^n(f_1\otimes f_2\cdots\otimes f_k) \to \int f_1\otimes\cdots\otimes f_k\ d\mu_k$$

in $L^2(\nu_k)$. This says that

$$\frac{1}{N_k-M_k}\sum_{n=M_k+1}^{N_k} T^nf_1\otimes T^{2n}f_2\otimes\cdots\otimes T^{kn}f_k \to \prod_{i=1}^{k}\int f_i\ d\mu$$

in $L^2(\nu_k)$. In light of the definition of ν_k,

$$\frac{1}{N_k-M_k}\sum_{n=M_k+1}^{N_k} f_1(T^nx)f_2(T^{2n}x)\dots f_k(T^{kn}x) \to \prod_{i=1}^{k}\int f_i\ d\mu$$

in $L^2(\mu)$. Multiply by $f_0(x)$ and integrate to find that

$$\frac{1}{N_k-M_k}\sum_{n=M_k+1}^{N_k}\int f_0(x)f_1(T^nx)\dots f_k(T^{kn}x)\ d\mu \to \prod_{i=0}^{k}\int f_i\ d\mu.$$

Replacing x by T^nx gives

$$\frac{1}{N_k-M_k}\sum_{n=M_k+1}^{N_k}\int f_0(T^nx)f_1(T^{2n}x)\dots f_k(T^{(k+1)n}x)\ d\mu \to \prod_{i=0}^{k}\int f_i\ d\mu,$$

that is

$$\frac{1}{N_k-M_k}\sum_{n=M_k+1}^{N_k}\int T_{k+1}^n(f_0\otimes f_1\otimes\cdots\otimes f_k)\ d\nu_{k+1} \to \int f_0\otimes f_1\otimes\cdots\otimes f_k\ d\mu_{k+1}.$$

□

For simplicity, we suppress subscripts k in the following corollaries.

Corollary 4.2. *If* $(X,\mathcal{B},\mu,T)$ *is weak mixing and* $A_1,A_2,\dots,A_k\in\mathcal{B}$ *then*

$$\lim_{N-M\to\infty}\frac{1}{N-M}\sum_{n=M+1}^{N}\mu(A_1\cap T^nA_2\cap\cdots\cap T^{(k-1)n}A_k) = \mu(A_1)\mu(A_2)\dots\mu(A_k).$$

Proof. Apply Theorem 4.1 to $f_i = \chi_{A_i}, i = 1, \ldots, k$, where each χ_{A_i} is the characteristic function of A_i. $\square$

Corollary 4.3. *If $(X, \mathcal{B}, \mu, T)$ is weak mixing and $A_1, A_2, \ldots, A_k \in \mathcal{B}$, then*

$$\lim_{N-M\to\infty} \frac{1}{N-M} \sum_{n=M+1}^{N} |\mu(A_1 \cap T^n A_2 \cap \cdots \cap T^{(k-1)n} A_k) - \mu(A_1)\mu(A_2)\ldots\mu(A_k)| = 0.$$

Proof. By applying Corollary 4.2 to the weak mixing transformation $T \times T$ on $X \times X$ and the sets $A_1 \times A_1, \ldots, A_k \times A_k$, we obtain

$$\frac{1}{N-M} \sum_{n=M+1}^{N} \mu \times \mu[(A_1 \times A_1) \cap (T \times T)^n (A_2 \times A_2) \cap \cdots \cap (T \times T)^{(k-1)n} (A_k \times A_k)]$$
$$\to (\mu \times \mu)(A_1 \times A_1)\ldots(\mu \times \mu)(A_k \times A_k),$$

that is

$$\frac{1}{N-M} \sum_{n=M+1}^{N} [\mu(A_1 \cap T^n A_2 \cap \cdots \cap T^{(k-1)n} A_k)]^2 \to [\mu(A_1)\mu(A_2)\ldots\mu(A_k)]^2.$$

If we let

$$a_n = \mu(A_1 \cap T^n A_2 \cap \cdots \cap T^{(k-1)n} A_k)$$

and

$$\alpha = \mu(A_1)\mu(A_2)\ldots\mu(A_k),$$

then

$$\frac{1}{N-M} \sum_{n=M+1}^{N} (a_n - \alpha)^2 = \frac{1}{N-M} \sum_{n=M+1}^{N} a_n^2 - 2\alpha \frac{1}{N-M} \sum_{n=M+1}^{N} a_n + \alpha^2 \to 0,$$

since the first term tends to α^2 and the second to $-2\alpha^2$. By the Cauchy-Schwarz Inequality, we obtain

$$\frac{1}{N-M} \sum_{n=M+1}^{N} |a_n - \alpha| \le \sqrt{\sum_{n=M+1}^{N} |a_n - \alpha|^2} \sqrt{\frac{1}{N-M}} \to 0,$$

as desired. $\square$

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